

A conditional lifting theorem for several small dice

Complete proof; independently reviewed

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Status. Full independent mathematical reviews by the root, rational-design, and independent-direction agents passed. This is not formal verification.

Scope. This is a conditional finiteization theorem. Its hypotheses are verified by the rational smoothing construction in `two_polynomial_exceptional_dice.tex`. For general k , the present note is only a conditional criterion.

Theorem 1. *Let $m, k \geq 1$. Suppose independent variables $A_1, \dots, A_k, U_1, \dots, U_m$ are fully permutation-fair, where each U_j is uniform on $[0, 1]$ and each A_i is uniform on K_i distinct interior points. Suppose their atom locations admit a partition into open intervals (clusters) such that:*

1. *all cluster closures are disjoint and contained in $(0, 1)$;*
2. *each cluster contains atoms of exactly one A_i ;*
3. *each cluster contains at least m^2 distinct atoms;*
4. *each cluster moment $K_i^{-1} \sum_{a \in C} a^r$ is rational for $0 \leq r \leq m$.*

Then a completely finite permutation-fair family exists with exactly K_i faces on its i th specified die, $1 \leq i \leq k$. The other m dice may be chosen to have one common finite face count. No quantitative bound on that common count is asserted.

The point of the hypothesis is the number of distinct nodes in each cluster. Replacing one atom by coalescing replicas does not justify it: the Vandermonde inverse then deteriorates, and small geometric perturbations need not yield small density corrections.

Fix a full order of all $m + k$ labels. Conditional on the k specified values $t_1, \dots, t_k$, its probability for uniform old variables is zero unless the specified labels have their prescribed relative order. In the correct chamber it is

$$P(t) = \prod_{a=0}^k \frac{(t_{a+1} - t_a)^{r_a}}{r_a!}, \quad t_0 = 0, \quad t_{k+1} = 1, \quad \sum_a r_a = m, \quad (1)$$

where the specified variables have been listed in that order. On a product of clusters its chamber is fixed, and P is a polynomial of total degree at most m (or identically zero).

Lemma 1 (Several boundary responses). *Around each specified node t_q , choose a small interval, all such intervals disjoint, and assign the node to at most one old label. For a node assigned to label j , choose a signed density h_q with*

$$\int_{x < t_q} Q(x) h_q(x) dx = Q(t_q), \quad \int_{x > t_q} Q(x) h_q(x) dx = -Q(t_q) \quad (Q \in \mathcal{P}_{m-1}).$$

Give old label j density $1 + \sum_{q \text{ assigned to } j} \delta_q h_q$. For every specified variable i , cluster C , and old label j , suppose

$$\frac{1}{K_i} \sum_{q \in C: q \text{ assigned to } j} \delta_q Q(t_q) = B_{i,C}(Q) \quad (Q \in \mathcal{P}_{m-1}), \quad (2)$$

with the same functional $B_{i,C}$ for every j . Let $E_{i,C} = K_i^{-1} \sum_{q \in C} \text{ev}_{t_q}$. Then the probability of the fixed full order is

$$\sum_{C_1, \dots, C_k} \left[\prod_{i=1}^k (E_{i,C_i} + B_{i,C_i} \partial_i) \right] P. \quad (3)$$

Here each operator acts only on its own coordinate, and the appropriate polynomial branch of (1) is used on each cluster product.

Proof. Expand the product of the old densities in the order integral. Each old variable contributes at most one bump. If the support of any selected bump is not cut by one of the selected specified values, then, after integrating the untouched uniform variables, its integrand is a polynomial of degree at most $m - 1$ throughout its whole support. The support has fixed order relative to all other selected supports and cut points. Its full moments vanish, so that term is zero.

Consequently a surviving term assigns its selected old variables injectively to selected specified cut points; in particular there are at most k bumps. The truncated evaluation identities replace each such old integration by evaluation at its cut, with a positive sign on the lower side and a negative sign on the upper side. They are precisely the boundary terms obtained by differentiating (1) once in each of those cut coordinates.

For clarity about collisions, a gap containing r prescribed old labels contributes $l^r/r!$. A derivative at its lower endpoint removes the first old label with a minus sign, and one at its upper endpoint removes the last old label with a plus sign. If both act and $r \geq 2$, they remove distinct labels and give the corresponding second derivative. If $r = 1$, the second derivative is zero: there is no term using the same old variable twice. If $r = 0$, even the first derivative is zero. Since each specified cut is differentiated at most once, these are all possible collisions. Different gaps factor independently. This proves the claimed mixed-derivative interpretation with its exact coefficients.

Now sum over node choices for each fixed assignment of distinct old labels to the cuts. On a cluster product all remaining expressions are polynomials of degree at most $m - 1$ in each differentiated coordinate. Equation (2) replaces every such sum by $B_{i,C}$, independently of the old label that was assigned to it. The undifferentiated coordinates give $E_{i,C}$. Summing over all subsets of cut coordinates gives (3). $\square$

Proof of Theorem 1. Choose rational cluster endpoints and disjoint rational support intervals around all baseline nodes, with each support inside its cluster. For every cluster choose m disjoint groups of m nodes, one group for each old label. Unused nodes are unassigned. Approximate all baseline nodes by distinct rational nodes t_q inside their fixed support intervals. Denote the original cluster functional by $E_{i,C}^0$ and the new one by $E_{i,C}$.

Both have the same mass. Therefore

$$B_{i,C}(Q) = E_{i,C}^0(JQ) - E_{i,C}(JQ), \quad Q \in \mathcal{P}_{m-1}, \quad (4)$$

is well defined, where JQ is any polynomial antiderivative. The rational cluster-moment hypothesis ensures that all $B_{i,C}$ are rational, although the baseline nodes themselves need not be. They tend to zero as the rational approximations approach the baseline nodes.

For each old label's group in each cluster, the evaluation map on $\mathcal{P}_{m-1}$ is an invertible Vandermonde matrix. Solve the rational linear system (2) for the δ_q . Because the original

nodes within every group are distinct, the inverse matrices stay bounded in a sufficiently small neighborhood of the baseline. Thus all δ_q tend to zero.

The required h_q can be chosen rational and uniformly bounded as t_q varies near its baseline node. Choose m disjoint rational subintervals to the left of that neighborhood and m to its right, all inside the fixed support. The moment matrices of their indicator functions are invertible: a polynomial of degree at most $m - 1$ whose integral vanishes on each of m disjoint intervals has a zero in each interval, hence is zero. Solve separately for truncated moments $Q \mapsto Q(t_q)$ and $Q \mapsto -Q(t_q)$. The fixed inverse matrices and continuous right sides give the uniform bound. Taking the rational approximation sufficiently close ensures that all old densities stay positive, for instance between $1/2$ and $3/2$.

By construction

$$(E_{i,C} + B_{i,C}\partial_i)P = E_{i,C}^0 P \quad (P \in \mathcal{P}_m).$$

The response lemma, successively in each coordinate on every cluster product, reproduces exactly the original order probability. Hence the new rational specified nodes and the positive rational step densities of the old variables are fully fair.

Finally partition at every density breakpoint and every specified node. All old interval masses are rational. Take any finite fair m -die block with F faces per die and a common denominator D for those masses. Within each interval replace the continuous old variables by this block, independently replicating each old letter by its integer interval-mass numerator. Conditional on which old variables fall in that interval, their order remains uniform. Each old die has exactly FD faces. Insert one specified face at each of its rational nodes and assign globally distinct integer labels. This preserves every complete order and finishes the proof with the stated face counts. $\square$

Unresolved hypothesis. For general k , this remains a conditional criterion. The companion two-variable smoothing proof supplies the hypotheses when $k = 2$. Rationality of the cluster moments cannot simply be dropped when the desired final model has equiprobable finite faces.